

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd term Examination, 2022

EE 1213

(Electrical Circuits and Electronics)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

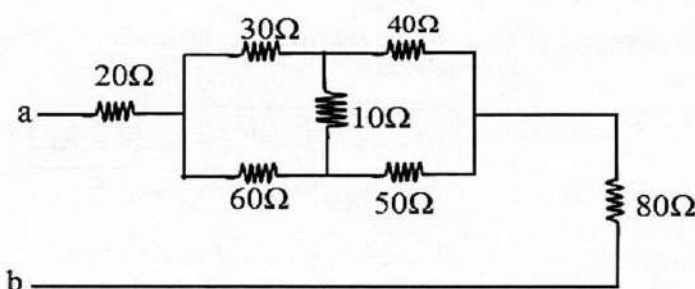
ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

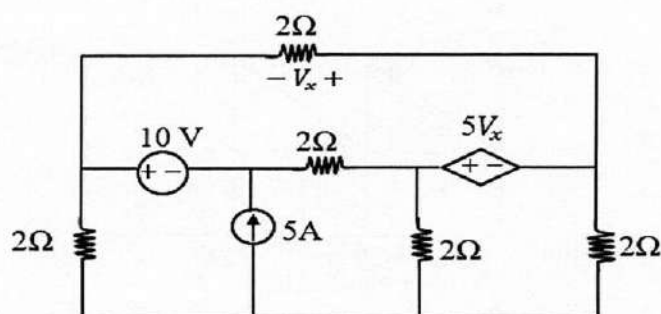
SECTION – A

1(a). Define active element and passive element. State and explain Kirchhoff's voltage and Kirchhoff's current law. 10

1(b). Obtain the equivalent resistance R_{ab} in the following circuit. 10

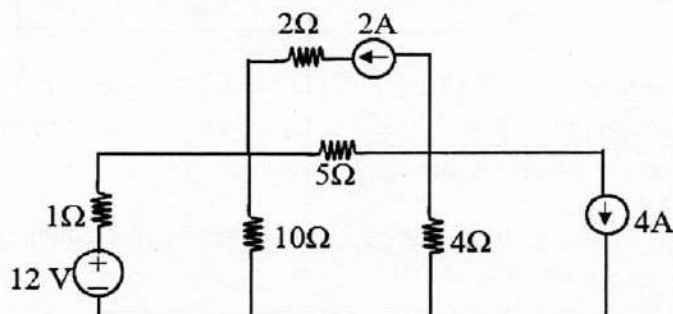


1(c). Define node, branch, and mesh. Find the node voltages in the following circuit. 15

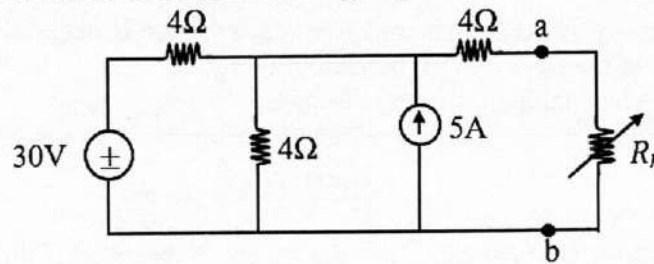


2(a). Briefly explain the terms supernode and supermesh. How to solve a circuit containing supermesh? 10

2(b). State the Superposition theorem. Calculate the voltage across 10Ω and current through 4Ω of the following network using Superposition theorem. 15



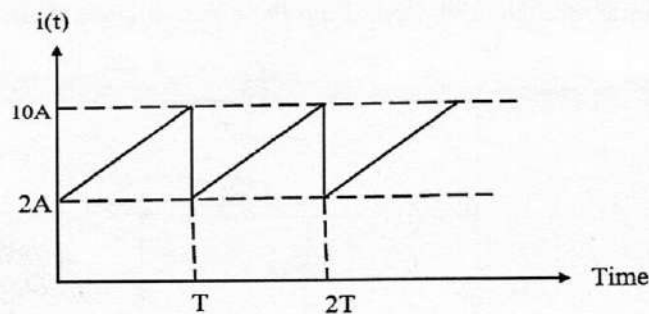
- 2(c). Find the Thevenin's equivalent circuit of the following circuit with respect the terminals **a-b**. Hence find the current through $R_L = 6, 16$ and 36Ω . 10



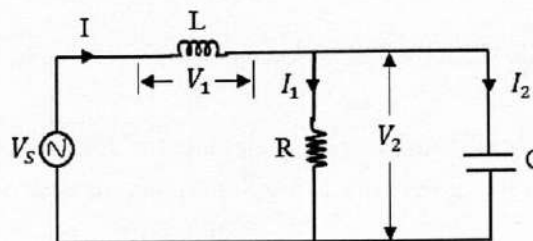
- 3(a). State the maximum power transfer theorem. Derive the condition for maximum power transfer and calculate the maximum power delivered to the load under that condition. 12
- 3(b) Find all the possible roots of the following expression: 10

$$\sqrt[5]{\frac{10\angle 45^\circ \cdot 5e^{j60^\circ}(-4.05 + j3.5)}{\log_e 50\angle 50^\circ(1 - j5)}}$$

- 3(c). What is the physical significance of RMS value of current in a *ac* circuit? Determine the form factor of the following waveform: 13



- 4(a). Draw the phasor diagram for the following circuit. 08



- 4(b). Two impedances $Z_1 = (10 + j15)\Omega$ and $Z_2 = (18 + j6)\Omega$ are connected in parallel across a voltage of $V = 200 + j0$. Calculate the total current, its phase and the total active power in the circuit. 15
- 4(c). Define resonance. Derive the expression of resonant frequency and Q-factor of a series resonance circuit. 12

SECTION – B

- 5(a). How the electron-hole pair is generated in semiconductor? Describe the importance of semiconductor doping. Then, Explain the n-type of doping Si with necessary diagrams. 13
- 5(b). State and derive mass action law. Also, mention the validity of this law. 13
- 5(c). Find the conductivity of a 1 cm^3 pure silicon crystal. Determine is the conductivity when the crystal is doped with arsenic if the doping is 1 *part per billion(ppb)*. Given that, atomic concentration in silicon is $5 \times 10^{22} \text{ cm}^{-3}$, $n_i = 1 \times 10^{10} \text{ cm}^{-3}$, $\mu_e = 1350 \text{ cm}^2 \text{v}^{-1} \text{s}^{-1}$, and $\mu_h = 450 \text{ cm}^2 \text{v}^{-1} \text{s}^{-1}$. 09
- 6(a). Discuss the dependence of electron concentration and conductivity of a n-type semiconductor on temperature with showing different temperature ranges. 11
- 6(b). What will happen if an intrinsic semiconductor has n-type doping on one side and p-type doping on the other side? Derive the built in voltage across a p-n junction. 13
- 6(c). Calculate the built in potential for a p^+n junction at 27°C having donor and acceptor concentration of 10^{16} cm^{-3} and $5 \times 10^{18} \text{ cm}^{-3}$, respectively. Also, calculate the depletion width. 11
- 7(a). Compare between avalanche breakdown and zener breakdown. Write down the characteristics and application of Schottky diode. 08
- 7(b). Draw the energy band diagram of a p-n junction under i) Open circuit, ii) Forward bias, and iii) Reverse bias. 09
- 7(c). An n-type Si wafer has been doped with 10^{16} cm^{-3} antimony(Sb) atoms. Calculate the position of the Fermi energy with respect to the Fermi Energy E_{Fi} in intrinsic Si. The above n-type Si is further doped with boron atoms of $2 \times 10^{17} \text{ cm}^{-3}$. Now, Calculate the position of the Fermi energy with respect to the E_{Fi} in intrinsic Si. Assume, $T=300\text{K}$. 09
- 7(d). Design a BCD counter using JK flip-flop. 09
- 8(a). Why NAND and NOR gates are called universal gate? Implement $F = (A + B')(CD+E)$ using NAND gates. 11
- 8(b). Define flip-flop, Latch, and clock pulse. What are the limitations of RS flip-flop and how to overcome them using JK flip-flop. Explain with necessary logic diagrams. 13
- 8(c). Define register. Discuss the operation of a 4-bit bidirection shift register with parallel load using necessary diagrams. 11

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2022

Hum 1213

(Technical English)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Make sentences with the following structures using the words given in brackets. 14
- (i) Subject + Intransitive verb + Adverb. (Live as verb)
 - (ii) Subject + Transitive verb + Object. (Write as verb)
 - (iii) Subject + Linking verb + Adjective Complement. (Seem as verb)
 - (iv) That + Subject + Verb + Adverb of Manner + Verb + Adjective complement. (Read and is as verb)
 - (v) Subject + Verb + Adverb of Manner + that + Subject + Verb + Object + Adverb. (Study and do as verb)
 - (vi) Subject + Relative pronoun + Verb + Adverb + Verb + Noun complement. (Work and is as verb)
 - (vii) Subject + Linking verb + not only + Adjective complement + but also + adjective complement. (Is as verb)
- 1(b). Change the following words as asked in brackets and make sentences with the changed forms. 12
- (i) Rarity (into adjective), (ii) Stable (into noun), (iii) Invade (into noun), (iv) Betray (into noun), (v) Pacification (into verb), and (vi) Bed (into verb).
- 1(c). Make use of the following words in sentence as directed. 09
- Over (adv.), Right (verb), Once (adv.), round (pre.), So (conj.), Watch (noun)
- 2(a). Transform the following sentences as asked in brackets. 14
- (i) That you can recite poetry well is praiseworthy. (Simple)
 - (ii) Habib, a banker, works sincerely in office. (Complex)
 - (iii) Liza can't reach the meeting in time because of her walking slowly. (Complex)
 - (iv) The beauties of nature are beyond description. (Interrogative)
 - (v) Nobody will deny that I am right. (Affirmative)
 - (vi) To add to his worries his enterprises failed. (Compund)
 - (vii) This is one of easiest subject. (Positive)
- 2(b). Make WH questions with the following sentences. 12
- (i) There was an accident.
 - (ii) He left the shop without a word.
 - (iii) There is no use of running so fast.
 - (iv) Between you two, you will accompany me.
 - (v) I don't find any logic in his statement.
 - (vi) It is very inconvenient to stay with me in such a small room.

- 2(c). Write one synonym and one antonym for each of the following words and make sentences with using the new words. 09
(i) Fictitious, (ii) Economise, and (iii) Honesty.
- 3(a). Make a new words with the following prefixes and suffixes, and use the new words in sentence. 09
Anbi —, By —, In —, — cy, — ce, — ing
- 3(b). Complete the following sentences with clauses as asked in brakets. 12
(i) _____ inspires us touchingly. (Noun clause)
(ii) The gentility in his character is _____. (Noun clause)
(iii) _____, you will try heart and soul. (Adv. clause)
(iv) _____, she achieves a good result in exam. (Adv. clause)
(v) _____, his parents can feel happy. (Adj. clause)
(vi) Shimu, _____, teaches in KUET. (Adj. clause)
- 3(c). Express the following notions/functions in sentences: 09
Negotiate, Refuse to buy, Suggest, Making a choice, Getting attention, Apology.
- 3(d). Make sentences with the following phrasal verb: 05
Turn down, take after, hold back, fall apart, mix up.
- 4(a). Correct the following sentences: 14
(i) Mim goes to the campus everyday.
(ii) Shila can play harmonium.
(iii) When he is responsible, he can succeed in life.
(iv) The teacher wants some blotting.
(v) The clock struck ten hours.
(vi) The car is out of repairs.
(vii) We have read the work of J. Keats.
- 4(b). Make use of the following modals in sentence as asked in braekets. 12
(i) Can. (To express ability)
(ii) Could. (To express past ability)
(iii) Might. (To express uncertainty)
(iv) May. (To express guess about the past)
(v) Should. (To express recommendation)
(vi) Should. (To express expectation)
- 4(c). Make sentences with the following phrases and idioms: 09
(i) Add insult to injury, (ii) Piece of cake, (iii) The elephant in the room, (iv) Clouds on the horizon, (v) Beat the bullet, and (vi) cold feet.

SECTION – B

- 5(a). Read the following passage carefully and answer the questions that follow. 20
- Human beings live in society. They are not independent, can't live alone and so usually communicate each other in the society. Here so attachment in a comfortable relation must stay in between one of a society with another one. Everyone of a society should account his/her neighbor's flesh, blood, honor, assets, feelings etc. as pure and intact as his/her own one's. So, fondness must be for other ones in any way and at any cost. Any excuse to share affection with other ones of a society can't work from humans, it must be obeyed. This strong will can't separate attachment from humans in society. This scenario enriches magnificently the structure of the life of a society. J Keats ss rightly says – 'Here (in the world) where men sit and hear each other groan.' ('Ode to a Nightingale'). As a result of lacking attachment, we perceive the heart broken scenario of life in the world. We are with the disquietness of the relation in brothers, sisters, neighbors, friends, colleagues etc. Consequently, the life of pains continues in society. We forget the idea 'I celebrate myself and sing myself/And what I assume you shall assume;' (Walt Whitmen). Here the poet merges himself with the self of others.
- (i) Why is attachment in society essential?
 - (ii) What of neighbors should one consider and how should one be so?
 - (iii) What can't separate attachment in society and what is the result?
 - (iv) As a result of not remembering one's self merging in others' self what happen in society?
- 5(b). Write a précis of the above passage mentioned in question no. 5(a) with a suitable title. 15
- 6(a). Write a cause and effect paragraph on using sarcasm in everyday conversation. 15
- 6(b). Amplify the idea contained in the following statement: 20
- "It is not what you know, it's who you know"
- 7(a). **Cobblestone Energy** is looking for an Energy Science Engineer to work in wholesale electricity markets in Dubai, UAE. Prepare your CV and apply for the post. 20
- 7(b). Suppose you are the new CEO of a company. Write a memo to host a retirement party for the ex-CEO. 15
8. Write a free composition on any one of the followings: 35
- (i) "Quality depravation in education". A destruction in a country development.
 - (ii) Lack of time management on daily productivity.

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2022

ESE 1205

(Thermodynamics for Energy Engineering)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). With proper example state zeroth law of thermodynamics. Which thermodynamic property is based on this law? 10
- 1(b). What is phase rule? Write down the phase rule for a non-reacting system. 04
- 1(c). Write short notes on the following terms: (i) Compressibility factor, (ii) Principle of corresponding state. 06
- 1(d). Determine the pressure of nitrogen gas at $T = 175\text{ K}$ and $v = 0.00375\text{ m}^3/\text{kg}$ on the basis of 15
(i) The ideal gas equation of state.
(ii) Soave-Redlich-Kwong equation of state.
- 2(a). “An energy transfer can be heat or work, depending on system selection”— explain the statement. 10
- 2(b). State the first law of thermodynamics. For an open system having one inlet and outlet, derive the mathematical form of first law of thermodynamics by considering the system has no interactions with its surroundings other than the mass flows and the associated changes in volume which result in work terms. 15
- 2(c). Air enters in a compressor operating at steady state at a pressure of 1 bar, a temperature of 290 K, and a velocity of 6 m/s through an inlet with an area of 0.1 m^2 . At the exit, the pressure is 7 bar, the temperature is 450 K and the velocity is 2 m/s. Heat transfer from the compressor to its surroundings occurs at a rate of 180 kJ/min. Employing the ideal gas model, calculate the power input to the compressor in kW. If the change in kinetic energy from inlet to exit were neglected, evaluate the compressor power in kW, keeping all the data unchanged. (Use 1 bar = 100 KPa) 10
- 3(a). State the Clausius and Kelvin-Planck statement of second law of thermodynamics. With schematic diagram, explain the equivalence between this two statements. 10
- 3(b). “The efficiency of an irreversible heat engine is always less than the efficiency of a reversible one operating between the same two reservoirs”— prove the statement. 08
- 3(c). Determine the compressor work input required to compress steam isentropically from 100 KPa to 1 MPa, assuming that the steam exists as (i) saturated liquid, and (ii) saturated vapor at the inlet state. 08
- 3(d). Explain the increase in entropy principle. 09
- 4(a). For ideal gas derive the following relation: $S_2 - S_1 = C_p \ln\left(\frac{v_2}{v_1}\right) + C_v \ln\left(\frac{p_2}{p_1}\right)$ 09
- 4(b). In the absence of any friction and other irreversibilities, can a heat engine have an efficiency of 100 percent? Explain. 04

- 4(c). What is exergy? Explain the physical significance of exergy. 04
- 4(d). A piston-cylinder device contains 0.05 Kg of steam at 1 MPa and 300 °C. Steam now expands to a final stage of 200 kPa and 150 °C, doing work. Heat losses from the system to the surroundings are estimated to be at $T_0 = 25\text{ °C}$ and $P_0 = 100\text{ kPa}$, determine: 18
- (i) The exergy of the steam at the initial and final states.
 - (ii) The exergy change of the steam.
 - (iii) The exergy destroyed and
 - (iv) The second law efficiency for the process.

SECTION – B

- 5(a). How can we increase the efficiency of the Rankine cycle? 10
- 5(b). Why regeneration is done in Rankine cycle? What is cogeneration? 07
- 5(c). Consider a steam power plant operating on the ideal refrigerative Rankine cycle with one open feed water heater. Steam enters the turbine at 15 MPa and 600 °C and is condensed in the condenser at a pressure of 10 KPa. Some steam leaves the turbine at a pressure of 1.2 MPa and enters the open feed water heater. Determine the fraction of steam extracted from the turbine and the thermal efficiency of the cycle. 18
- 6(a). What is air standard cycle? What are the assumptions associated with this cycle? 08
- 6(b). With a neat schematic diagram, describe the working principle of Ammonia-Water absorption refrigeration system. 12
- 6(c). A gas turbine working on an air standard Brayton cycle operates between the temperature limits of 300 K and 1200 K and pressure limits of 101 kPa and 505 kPa. Calculate: 15
- (i) Thermal efficiency of the cycle, (ii) Compressor work in kJ/kg, and (iii) Turbine work in kJ/kg.
- 7(a). Do diesel or petrol engines operate at higher compression ratios? Why? 10
- 7(b). State Dalton's law of partial pressure. What is meant by gravimetric analysis and volumetric analysis? 09
- 7(c). A mixture of ideal gases consists of 5 Kg of nitrogen and 6 Kg of carbon dioxide at a pressure of 4 bar and a temperature of 27 °C. Find: (i) The mole fraction of each constituent, (ii) The equivalent molecular weight of the mixture, (iii) The equivalent gas constant of the mixture, and (iv) The volume and density of the mixture. 16
- 8(a). "A cooling effect cannot be achieved by throttling unless the fluid is below its maximum inversion temperature"— explain the statement. 10
- 8(b). What is Maxwell relation? Why do we need Maxwell relations? Express the Maxwell relations in equation form. 13
- 8(c). Atmospheric air at 1.0132 bar has 32 °C DBT and 26 °C WBT. Calculate: (i) Partial pressure of water vapor, (ii) Relative humidity, (iii) Specific humidity, and (iv) Enthalpy of the mixture. 12

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2022

Math 1213

(Differential Equation and Coordinate Geometry)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Define invariants. Find the transformed coordinates of $(3, -1)$ with respect to the new rectangular axes of x and y , whose equations are $-3x + 4y + 3 = 0$ and $4x + 3y - 4 = 0$, respectively. 09
- 1(b). Find the transformed equation of $4xy - 3y^2 = c^2$ when the rectangular axes are rotated through an angle $\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$. 12
- 1(c). Identify the equation of curve, $x^2 - 4xy + y^2 + 8x + 2y - 5 = 0$ and reduce it to the standard form. 14
- 2(a). Find the cartesian and cylindrical polar coordinates for the point whose spherical polar coordinates are $\left(5, \frac{\pi}{4}, \frac{3\pi}{4}\right)$. 12
- 2(b). Define direction cosines of a line. Prove that $l^2 + m^2 + n^2 = 1$, where l, m, n are the direction cosines of a line. 13
- 2(c). Find the coordinates of the point where the join of $(2, -3, 7)$ and $(-3, 2, 2)$ intersects xz plane. 10
- 3(a). What is meant by plane? Find the equation of a plane which passes through the point $(1, 2, 3)$ and perpendicular to the line joining the points $(5, 4, -2)$ and $(9, -3, 1)$. 12
- 3(b). A plane passes through the points $P(1, 2, -1)$, $Q(2, 3, 1)$, $R(3, -1, 2)$. Find the normal vector of this plane and hence using this normal, find the equation of this plane. 12
- 3(c). Find the angle between the objects represented by $\frac{x+1}{2} = \frac{y}{3} = \frac{z-3}{6}$, and $10x + 2y - 11z = 3$ using vector method. 11
- 4(a). Find the vector equation as well as parametric equation of a line passing through the points $(-1, 0, 2)$ and $(3, 4, 6)$. 10
- 4(b). Find the shortest distance between the two skew-lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-4}{4}$ and $\frac{x-2}{2} = \frac{y-4}{4} = \frac{z-5}{5}$. 13
- 4(c). Test the four points $(2, 3, 2)$, $(0, 1, 2)$, $(4, 3, 6)$, and $(3, 0, 1)$ are coplanar or not. If they are coplanar, then find the volume of a tetrahedron whose vertices are these four points. 12

SECTION – B

- 5(a). Define differential equations and classify them based on their type, order, and linearity. Also give example for each of the category. 10
- 5(b). What is the basic difference between the general solution and particular solution of a differential equation? Find the general solution of $(x + 2)^2 \frac{dy}{dx} = 5 - 8y - 4xy$. 12
- 5(c). Define homogeneous function of two variables. Identify whether the equation $xydx + (2x^2 + 3y^2 - 20)dy = 0$ is homogeneous, exact or not. Hence solve it accordingly. 13
- 6(a). Define initial value problem with an example. Solve the D. E. $\frac{d^2y}{dx^2} + 4y = 0$,
Where, $y(0) = 1$, and $y\left(\frac{\pi}{2}\right) = 2$. 12
- 6(b). Solve the D.E. $3(x + y)dx - xdy = 0$ 10
- 6(c). Solve: $(D^2 - 4D + 3)y = 3e^x \cos(2x) + x + 1$. 13
- 7(a). Define homogeneous linear differential equation of order “n” with an example. 04
- 7(b). Initially 100 mg of a radioactive substance was present. After 6 hours the mass had decreased by 3%. If the rate of decay is proportional to the amount of substance present at time t , find the amount of substance remained after 24 hours. 14
- 7(c). Solve $y'' + y' - 2y = xe^x$ using the method of variation of parameters. 17
- 8(a). Find the Laplace transform of the function 10
- $$F(x) = \begin{cases} \sin(x) & \text{for } 0 < x < \pi \\ 0 & \text{for } \pi < x < 2\pi \end{cases}$$
- 8(b). Find the inverse Laplace transform of $F(s) = \frac{s^2}{(s^2+4)^2}$ using Convolution theorem. 12
- 8(c). Solve: $y'' + 16y = f(t)$; $y(0) = 0, y'(0) = 1$ by using Laplace transform. 13

$$\text{Where, } f(t) = \begin{cases} \cos(4t); & 0 \leq t < \pi \\ 0 & t \geq \pi \end{cases}$$

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2022

Ch 1213

(Chemistry II)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

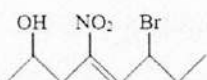
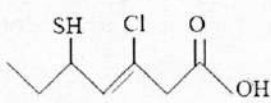
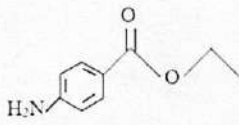
ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What is meant by unimolecular and pseudo-first order reactions? Provide example. 09
- 1(b). Why high molecularity reactions are rare? Explain with diagram. 08
- 1(c). What is activated complex? Discuss the collision theory of reaction rate. 10
- 1(d). The decomposition of C_2H_6 to methyl radicals is a first order reaction with a rate constant of $5.36 \times 10^{-4} S^{-1}$ at $700^\circ C$. Calculate the half-life of the following reaction in minutes. 08
- $$C_2H_6(g) \rightarrow 2CH_3(g)$$
- 2(a). Surface property is different than that of bulk properties – Interpret the statement. 09
- 2(b). Discuss the factors which affect the adsorption of a gas or liquid on a solid adsorbent. 08
- 2(c). Write down the differences between physical adsorption and chemical adsorption. 08
- 2(d). Deduce Langmuir adsorption isotherm equation and discuss this equation for the limiting conditions of very low and very high pressure. 10
- 3(a). Define photoexcitation. What could be the fate of photoexcited molecules? 07
- 3(b). How do photochemical reactions differ from thermal reactions? 08
- 3(c). What is quantum yield? Why photochemical reactions of H_2 and Cl_2 is a high quantum yield reaction? 10
- 3(d). Distinguish between fluorescence and phosphorescence. Draw energy level diagram to explain the process and discuss them briefly. 10
- 4(a). Mention the Ostwald classification of catalytic reaction. 08
- 4(b). Suggest why the rate of enzyme catalyzed reaction is maximum at optimum temperature. 09
- 4(c). Discuss the intermediate compound formation theory of catalysis. 10
- 4(d). Indicate some commercially important reactions that employ heterogeneous catalysts. 08

SECTION – B

- 5(a). Neopentane is gas while n-pentane is liquid – articulate based on London dispersion forces. 08
- 5(b). The molecular formula of a compound is $C_4H_{10}O$. 10
- (i) Draw the bond-line formula for all constitutional isomers.
 - (ii) Depict the orbital picture of a secondary and tertiary isomer.
 - (iii) Which isomer has higher melting point and why?
- 5(c). The stability of trans compounds outweighs the stability of cis compounds – construe in terms of heat of reaction. 08
- 5(d). Draw the condensed formula and provide IUPAC nomenclature of the following compounds: 09
- (i) 
 - (ii) 
 - (iii) 
- 6(a). Write down the vibrational energy expression of a diatomic molecule by indicating the meaning of each term therein. Sketch the energy levels and transition between them. 09
- 6(b). What is meant by allowed transitions and forbidden transitions? Describe the various types of absorption band which arise as a result of the electronic transition. 08
- 6(c). Rank the following in their increasing order of wave number for carbonyl ($-C=O$) group. Give reasons. 09
- $HCHO$, CH_3COCH_3 , CH_3CHO , CH_3CH_2CHO
- 6(d). The spectra of HCl displays a very intense absorption at 2886 cm^{-1} , a weaker one at 5668 cm^{-1} and a very weak one at 8347 cm^{-1} . Write the names of these peaks. Calculate the equilibrium oscillation frequency ($\bar{\omega}_e$), anharmonicity constant (x_e) and force constant (k). 09
- 7(a). What are the bathochromic and hypsochromic shifts? In polar solvents $n \rightarrow \pi^*$ transitions undergo bathochromic shift but $\pi \rightarrow \pi^*$ transitions usually undergo hypsochromic shift. Explain. 08
- 7(b). Absorption bands of UV-visible spectra are generally broad in nature – comment on the statement. 09
- 7(c). Graphically analyze the different sort of shifts that is observed in UV-visible spectra. 10
- 7(d). How are molecules categorized on the basis of moment of inertia? Classify the following molecules accordingly: 08
- OCS , CH_3Cl , CH_4 , H_2S , BCl_3 and SF_6 .
- 8(a). What is meant by passivity of metal? Explain the active-passive behavior of metal with diagram. 10
- 8(b). Articulate the dependency of corrosion rate on several parameters for underground corrosion. 08
- 8(c). Cite the role of temperature on under water corrosion with neat sketch. 09
- 8(d). Buried pipe can be prevented from corrosion by anodic protection – justify the statement with reaction and diagram. 08

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